SYSDESIGN REFERENCE

Caching Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Trade controlled staleness and invalidation complexity for lower latency and origin load.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / CACHING

SYSDESIGN

Caching Tradeoff Matrix

CORE IDEA

Trade controlled staleness and invalidation complexity for lower latency and origin load.

Reference architecture decision chain

01	02	03	04
Placement	Key Design	Ttl	Eviction

Decisions to defend

- Placement
- key design
- TTL
- eviction
- consistency
- and cache-aside vs write-through.

Failure modes to test

- Hot keys
- stampedes
- stale data
- low hit rate
- and oversized values.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Caching Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

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